

Predicting Team Success From Formation and Personnel in NCAA Division I Soccer

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Abstract

Predicting performance is a cornerstone of sports analytics, and in soccer much of that performance comes down to tactics: which formation a team plays and which players fill each role. We study whether these choices can be predicted from data by modeling the expected goal (xG) differential a team produces within a half. We compare several ways to encode a lineup, including data-driven player archetypes, season statistics, opponent information, and graph versus tabular representations. Across sixteen model variants, a graph neural network that combines data-driven archetypes with season statistics gave the best result, predicting xG differential with an R^2 of 0.283 and a sign accuracy of 0.659. These results suggest that team success can be partially predicted from formation and personnel choices alone.

1 Introduction

1.1 Problem and motivation

Predicting success within a soccer match is difficult for several reasons. In NCAA soccer, teams are allowed up to six substitution windows per match, so the set of players on the pitch changes many times within a match and within each half. To account for this, we model the difference in xG within a half rather than across a full match. Modeling at the half level lets the formation and personnel changes that happen at halftime appear as separate rows in the dataset.

The aim of this prediction task is to learn whether certain play styles across positions perform well together under given tactical conditions. A model of this kind could be used to simulate how a team’s expected performance changes when a player switches positions, or whether a player a team is recruiting fits tactically into the existing squad.

1.2 Research questions

Our goal is to determine whether specific tactical encoding choices help predict success. We break this into four questions:

1. **Personnel encoding.** Do data-driven player spatial archetypes beat raw position labels?
2. **Quality encoding.** Does a player’s season statistical profile improve prediction?
3. **Opponent conditioning.** Does adding the opponent’s lineup and personnel improve prediction (matchup vs. ego)?
4. **Representation.** Does a graph neural network outperform a tabular XGBoost model?

2 Related Work

2.1 xG modeling

Expected goals (xG) is the standard measure of shot quality in soccer. Early models estimate the probability that a shot becomes a goal from features such as shot location, body part, and defensive pressure, and show that spatial and contextual

features predict goals better than distance alone [1]. Later models improve these estimates using richer positional inputs and neural networks [2]. These models work at the level of individual shots: each shot is scored on its own, and a match’s xG is the sum over its shots. Our task is different. Instead of scoring shots, we predict the total xG differential a team produces over part of a match from the players on the pitch and how they are arranged, so the prediction reflects formation and personnel rather than the shots that happened to be taken.

2.2 Graph neural networks for soccer

Graph neural networks suit team sports because the useful information lies in how players relate to each other rather than in any single player. In soccer, players are represented as nodes and tactical or spatial relationships as edges, and message passing is used to predict outcomes such as match results or the value of a play [3]. Many of these models use graph attention [4] and the PyTorch Geometric library [5]. We build on this work but add features specific to within-match tactics: edge types for substitutions and for cross-team matchups based on shared time on the pitch, attention over a fully connected within-team graph rather than a fixed proximity graph, and node features that combine learned positional archetypes with season statistics.

2.3 Player role and archetype clustering

Other work replaces fixed position labels with roles learned from data. Bialkowski et al. learn team formations and player roles directly from tracking data and find that the learned roles match tactical intuition without supervision [6]. Related work describes individual play style using feature spaces built from large public event-data sets, so players can be compared by how they play rather than by their listed position. We take the same data-driven view but at a finer level: instead of one role per player, we cluster event-level spatial tendencies within each position group to form archetypes that describe different styles of playing the same position, and we use these archetypes as node features in place of raw position labels.

3 Methods

3.1 Problem definition

We treat each half of a match as one prediction example. For a half, let team 1 and team 2 be the two teams, and let xG_1 and xG_2 be the expected goals each team accumulates during that half. These come from Wyscout’s built-in xG values, summed over each team’s shots in the half. The target is the xG differential

$$y = xG_1 - xG_2.$$

The input is a snapshot of the half: the formation each team played, the players who appeared and the minutes they were on the pitch, each player’s raw position and assigned archetype, and season-level statistics for each player. A model takes this snapshot and predicts $\hat{y}$, an estimate of the half’s xG differential. A positive y means team 1 was expected to outscore team 2, so a model that recovers the sign of y has identified which team was expected to do better.

3.2 Data

The data comes from Wyscout [?] and covers more than 7,000 NCAA Division I matches from 2020 to 2024. Each match has event data, which records every on-ball event such as passes, shots, and duels, and formation data, which records each team’s lineup and formation.

We use the event data in two ways: to build the player archetypes used as model features, and to aggregate event outcomes into season statistics that are also used as features. The formation data is used to compute minutes played for each player and to build the half snapshots from the substitutions that occur during the match.

3.3 Player archetypes

3.3.1 Position groups

We first group Wyscout’s raw position labels into eight categories: center defenders (CD), left wide defenders (LWD), right wide defenders (RWD), central midfielders (CM), center forwards (CF), left wide players (LWP), right wide players (RWP), and goalkeepers (GK). These groups capture the left/right and front/back asymmetry of player roles. Within each group we then define styles of play,

so that two players listed at the same position can still play it in different ways.

3.3.2 Intra-event clustering

To define styles of play within a position group, we cluster each event type separately. For example, for center defenders we form 12 clusters of passes using the pass start and end coordinates. Each cluster corresponds to a common type of pass, and the share of a player’s passes that fall into each cluster describes part of their style. We repeat this for every event type and position group, which gives a set of clusters that any player’s events can be grouped into. Figure 1 shows the clusters for center defenders.

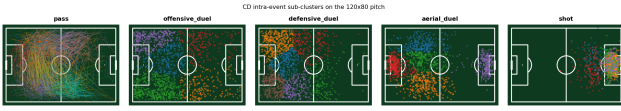


Figure 1: Intra-event k -means sub-clusters for center defenders. Each panel is one center-defender event type, with points colored by cluster. (All-groups version: Appendix A.)

3.3.3 Player-season aggregation

We then aggregate, for each player and season, the events that fall into each cluster. From these counts we build a vector that represents the player’s spatial tendencies: the proportion of each event type they perform, and how those events are distributed across the clusters.

3.3.4 Archetype clustering

Finally, we run k -means on the player-season spatial vectors to group players with similar tendencies into archetypes, separately for each position group. These archetypes are the features used by the models. Figure 2 shows the center-defender archetypes and how their spatial tendencies differ.

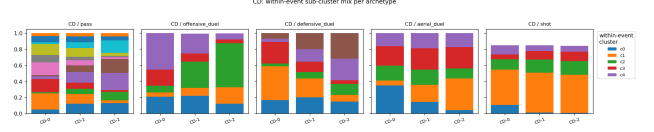


Figure 2: Within-event sub-cluster mix per archetype for center defenders. Each panel is one event type; stacked bars give the mean proportion of each archetype’s events that fall into each within-event sub-cluster. (All-groups version: Appendix B.)

3.4 Dataset construction

We build one snapshot per half. A snapshot records the formation a team played, every player who appeared in the half with the minute they entered and left the pitch, and each player’s archetype and raw position. We also attach each player’s season statistics. The target is the xG differential between the two teams in that half.

We split the data 80/20 into train and test sets, blocked by match so that both halves of a match fall in the same split. Every model variant is trained on the same training set and evaluated on the same held-out test set, so the results are directly comparable.

3.5 Models

3.5.1 Tabular XGBoost baseline

The baseline is a gradient-boosted tree model [?] trained on the dataset in its tabular form. It learns the relationship between the lineup features and the target without any explicit graph structure.

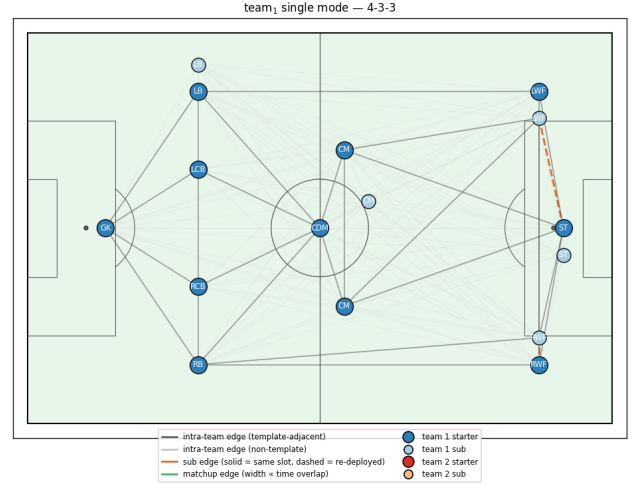
3.5.2 Graph neural network

The graph model takes the lineup for each half and builds a graph from it. A graph matches how soccer lineups are normally structured, drawn, and reasoned about: players are nodes and the relationships between them are edges, which lets the model represent how players interact and contribute to the team. The architecture is as follows:

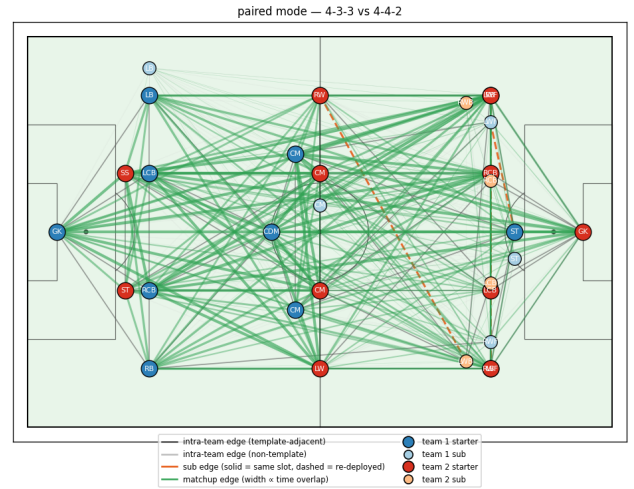
- **Node set.** Variable: the 11 starters plus any substitutes who appeared in the half (typically 3–6). Players are placed in a hand-crafted template based on the team’s formation, matching how lineups are usually drawn.

- **Node features.** Position encoding (raw position or archetype), season statistics, minutes played, and position within the formation template.
- **Edge types.**
 1. *Intra-team*: relationships between teammates.
 2. *Sub*: substitution information.
 3. *Matchup*: how a player relates to an opposing player.
- **Edge features.** A same-team flag, the relative distance between positions in the formation, and, for matchup edges, the differences in player statistics (for example, a strong winger against a weak defender).

Figure ?? shows both graph construction modes.



(a) **Single mode.** Team₁'s self-contained subgraph (a 4–3–3).



(b) **Paired mode.** Both teams in one 22+-node graph.

3.5.3 Model variants

Each model family has $2 \times 2 \times 2$ variants along three axes of tactical encoding:

- **Personnel encoding.** *position* (raw position labels) vs. *archetype* (data-driven clusters).
- **Opponent conditioning.** *ego* (single-team graph) vs. *matchup* (paired-team graph).
- **Quality encoding.** *no_stats* (no season stats) vs. *stats* (10-D player-season stats, plus stat-diff matchup features).

kind	opponent	stats	family	MAE	RMSE	MSE	R ²	sign acc
arch	ego	no	gnn	0.3415	0.4611	0.2126	0.1848	0.6174
arch	ego	no	tab	0.3673	0.4982	0.2482	0.0483	0.5838
arch	ego	yes	gnn	0.3217	0.4362	0.1902	0.2706	0.6519
arch	ego	yes	tab	0.3538	0.4793	0.2298	0.1191	0.5973
arch	matchup	no	gnn	0.3419	0.4640	0.2153	0.1744	0.6067
arch	matchup	no	tab	0.3446	0.4689	0.2198	0.1572	0.6151
arch	matchup	yes	gnn	0.3202	0.4326	0.1871	0.2826	0.6592
arch	matchup	yes	tab	0.3246	0.4415	0.1949	0.2526	0.6467
pos	ego	no	gnn	0.3623	0.5034	0.2534	0.0285	0.5459
pos	ego	no	tab	0.3918	0.5275	0.2782	-0.0668	0.5397
pos	ego	yes	gnn	0.3276	0.4437	0.1969	0.2450	0.6352
pos	ego	yes	tab	0.3507	0.4733	0.2240	0.1410	0.6028
pos	matchup	no	gnn	0.3648	0.5073	0.2574	0.0132	0.5326
pos	matchup	no	tab	0.3938	0.5246	0.2752	-0.0552	0.5825
pos	matchup	yes	gnn	0.3269	0.4431	0.1963	0.2473	0.6498
pos	matchup	yes	tab	0.3274	0.4407	0.1942	0.2555	0.6433

Design axis	MAE	RMSE	MSE	R ²	sign acc
archetype - position	-0.0162	-0.0227	-0.0222	+0.0851	+0.0308
stats - no-stats	-0.0319	-0.0456	-0.0433	+0.1662	+0.0578
matchup - ego	-0.0091	-0.0125	-0.0116	+0.0446	+0.0202
gnn - tab	-0.0184	-0.0203	-0.0194	+0.0744	+0.0110

4 Results

4.1 Model performance

The best model reaches an R² of 0.283 and a sign accuracy of 0.659, meaning it explains about 28% of the variance in xG differential and correctly calls which team was expected to outperform the other in roughly 66% of halves. This model uses every tactical encoding we tested: data-driven archetypes, the matchup graph, and season statistics. The encodings stack, with each one adding to predictive performance rather than replacing another.

Season statistics matter most. Adding them raises R² by 0.166 and lowers MSE by 0.043, far more than any other encoding. Season statistics capture the difference in player quality across lineups, which is a large part of what decides a half. They are also the easiest encoding to add, since they only require aggregated season totals and no

event data. That makes them a practical option for analysts who want a similar model but cannot build data-driven archetypes.

The archetypes also help, raising R² by 0.085 and lowering MSE by 0.022. Knowing how a player tends to play their position, not just which position they play, improves the prediction, and the model can use the mix of archetypes in a lineup. Archetypes cost more to build than season statistics because they need event data, but they add a clear gain on top of the statistics.

How the lineup is represented also matters. The graph form lets the model use the structure of the lineup and the relationships between players directly. The GNN raises R² by 0.074 and lowers MSE by 0.019 over the tabular baseline, so the graph captures lineup information that the tabular model misses.

The opponent conditioning helps least, raising R² by only 0.045 and lowering MSE by 0.011. Most

of a team’s expected output seems to come from its own tactics and player quality rather than the specific opponent. Opponent information is not useless, but it appears to add value mainly once a team’s own choices are accounted for, so teams are better off optimizing their own lineup before tailoring it to an opponent.

4.2 Limitations

A major limitation is that college soccer allows many substitutions, which adds noise to the lineup even within a half and makes it harder for the model to learn the relationship between the lineup and the xG differential. The model may perform better in professional leagues, where there are fewer substitutions and more tactical consistency within a match.

A second limitation is that the data spans every conference in Division I, where team and player quality vary widely. This variation adds noise. Normalizing player statistics across conferences could reduce it and let the model focus on the tactical relationships rather than raw quality differences.

A third limitation is the absence of match-level features such as home or away and travel distance. Home advantage is real in college soccer, and pitch type (grass versus turf) and weather vary across matches; teams unused to certain conditions may underperform. College athletes are also students, so heavy travel for away games can cause fatigue and less preparation time, which professional teams do not face to the same degree. None of this is currently captured.

A final limitation is that the archetypes require event data, which not every analyst can access. Our results show that season statistics alone already give useful predictions, but the archetypes add a clear gain on top of them, so there is value in building archetypes where event data is available.

4.3 Future work

One main use of this model is to simulate how different players would perform in a team’s current lineup. This would let us estimate how a new signing or a position change would affect expected performance, which could help coaches and analysts make recruitment and selection decisions.

A second direction is to learn the archetypes end-to-end inside the xG model rather than build-

ing them separately. Letting the model learn the spatial tendencies most useful for predicting xG differential could improve performance further.

A third direction is to apply the model to professional leagues, where there are fewer substitutions and more tactical consistency. This would test whether the same encodings work in a different setting, and whether the importance of each encoding changes across leagues.

5 Conclusion

This study looks at predicting team success from tactical lineup encodings, a task that matters for selecting game-day rosters and recruiting players. We encoded formations and personnel in several ways and described a data-driven method for building player spatial archetypes within position groups, which separates players by how they play their position. We then used graph neural networks to learn which styles of play work well together. The best model, a graph neural network using archetypes, player statistics, and opponent conditioning, predicted xG differential with an R^2 of 0.283. The approach is a promising starting point for predicting team success from formation and personnel.

Data and Code Availability

The match and event data are from Wyscout. The code used to build the archetypes, construct the dataset, and train and evaluate the models is available from the author at <https://github.com/zcochran27/team-success-prediction-via-formation-and-personnel>.

References

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- [3] P. Xenopoulos and C. Silva. “Graph Neural Networks to Predict Sports Outcomes.” *IEEE International Conference on Big Data (Big Data)*, 2021.
 - [4] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. “Graph Attention Networks.” *International Conference on Learning Representations (ICLR)*, 2018.
 - [5] M. Fey and J. E. Lenssen. “Fast Graph Representation Learning with PyTorch Geometric.” *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.
 - [6] A. Bialkowski, P. Lucey, P. Carr, Y. Yue, S. Sridharan, and I. Matthews. “Identifying Team Style in Soccer using Formations Learned from Spatiotemporal Tracking Data.” *IEEE International Conference on Data Mining Workshops (ICDMW)*, 2014.

A Intra-event sub-clusters, all position groups

The all-groups version of Figure 1: each row is one position group, each column is an event type. Where a group has fewer event types than the maximum, trailing panels are blank. The k -selection summary (elbow + silhouette per pair) is in the `event_clustering` notebook outputs.

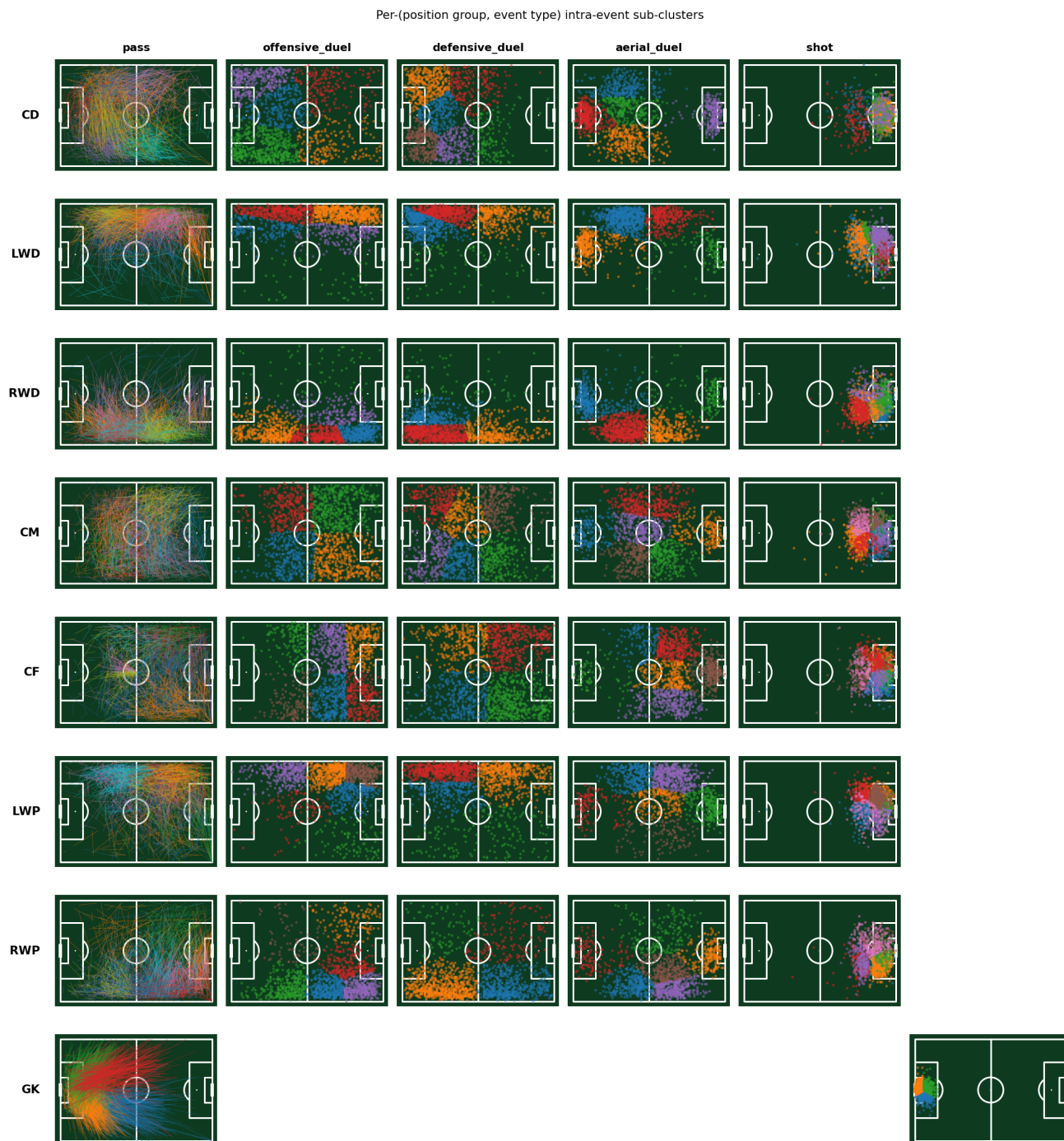


Figure 4: Intra-event sub-clusters for every (position group, event type) pair. Compare any single row against Figure 1 (the CD row).

B Within-event sub-cluster mix per archetype, all groups

The all-groups version of Figure 2: each row is one position group, each column is an event type, stacked bars give the proportional sub-cluster mix of that group's archetypes.

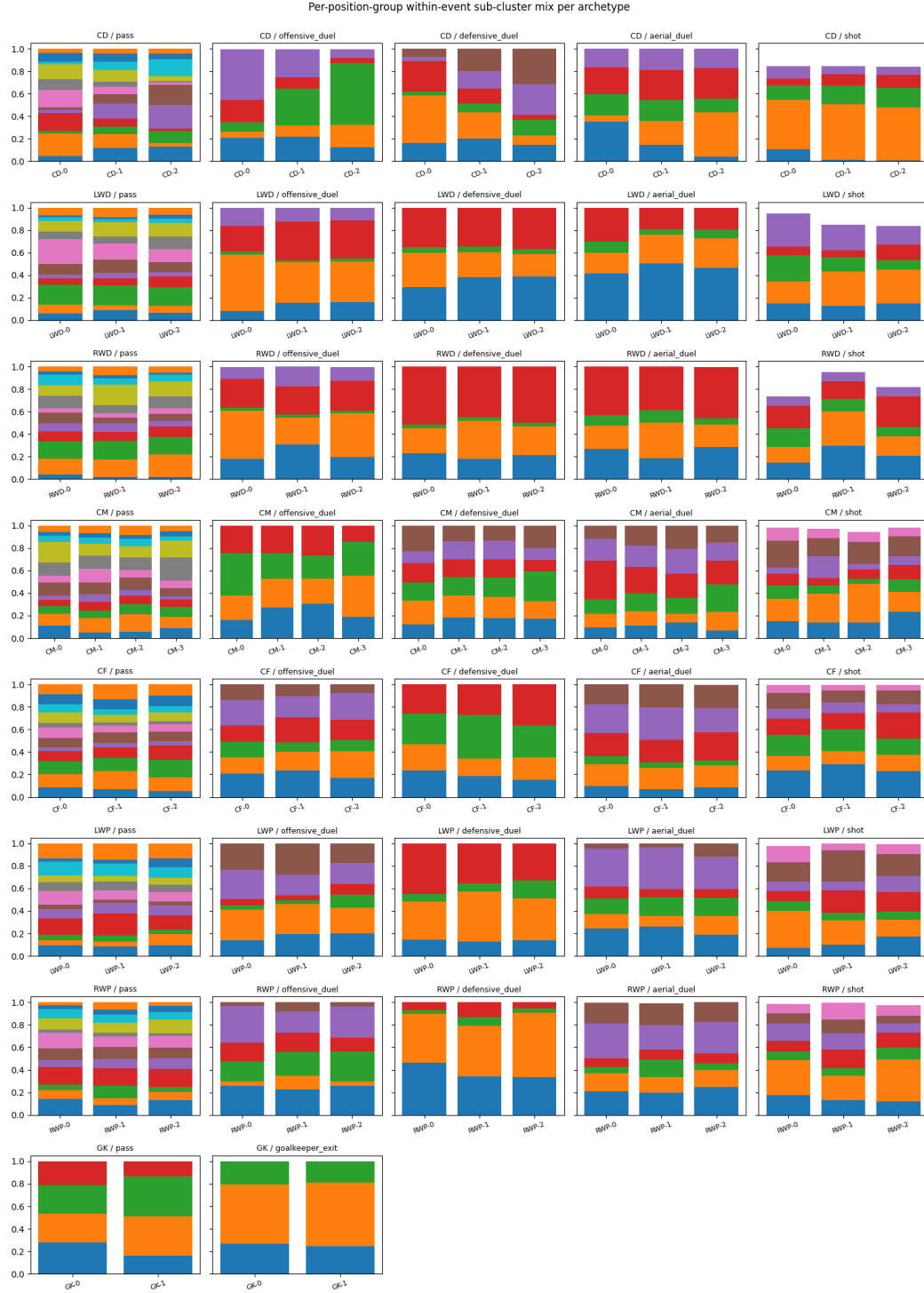


Figure 5: Within-event sub-cluster mix per archetype, every position group. Compare any single row against Figure 2 (the CD row).